

Integrating human-nature life support systems in the periphery of urban areas						Phase				
No.	Domain	Acronym	Type	Science Contribution	Innovation	Concept	Development	Prototype	Technical Skills	Associated with
1	Environmental Planning	GCC	Earth Science Model	1- Ensures high-quality training data and enables continuous monitoring of regional land change dynamics for spatiotemporal analysis and applied science decision-making.	1-Fused values of forest structure circumvent label interpretation biases.				Multisource remote sensing data pipelines, Open-sourced satellite image change detection monitoring for Earth-human coupled climate science, environmental science, carbon accounting, land change monitoring, ocean health analysis, humanitarian and disaster resilience planning, and applied science decision-making.	PhD Candidate at Tohoku University, Japan
					2-Spectro-vertical sample supervision increments data reliability.		✓			
2	Demographic Planning	Finecast	Population Downscaling Model	1- This model is particularly useful to evaluate population movement and projections in small tracts of land. It helps in forecasting, planning, and understanding spatiotemporal demographics for environmental science, humanitarian aid, risk management, disaster response, and recovery.	3-Regional metrics for high-quality training data collections.				Multi-scale dasymetric modeling, downscaling tabular population/mortality/fertility/migration data, data reliability estimation, optimized search and rescue operations, disaster shock estimation, demographic trend analysis, long-term recovery,	PhD Candidate at Tohoku University, Japan
					1- A Normalization method is proposed to reduce uncertainty derived from downscaling unprecise, census-based population data. This process circumvents the Modifiable Areal Unit Problem (MAUP) by offsetting scale variations generated by outlying data, enabling iterative dasymetric modeling to achieve very-fine-grained population distribution maps.					
3	Transportation Planning	OLUTM	Land Use and Transportation Model System	1- A significant probability of residential to mixed land-use change was found for middle-aged farmers with risk-taking lifestyles and upper median incomes venturing home-business entrepreneurs (HBEs). HBEs diversify rural livelihoods with non-farm activities that capitalize on local products and complement rural labor. This is important to sustain the environmental stability of rural areas. 2- A land-use treatment in nodes incited land-use conversion between the nodes. This is not common in land-use modeling research because it targets HBEs in an agronomic-centric network.	2-Unlike other models, this model was designed to allocate primary activities in undeveloped or developing areas and simulate the attraction of people from various scales of a city. The model provides a 10-year land development solution for a land development problem using 10 interactive models distributed into 2 integration processes with 11 environmental and socioeconomic indicators to evaluate change, and inform cross-sector policy strategies for climate change mitigation and sustainable economic development.	✓			hierarchies, activity-based scheduling, travel behavior modeling, micro-data, synthetic datasets, Monte Carlo microsimulation, multi-modal mobility, fuel consumption, carbon emissions, travel expenditures, population projections, redistribution, road volume capacity, street enclosure ratios, built capacity, local multiplier effects, shift-share approach, agglomeration economies, bid-rent theory of urban accessibility, econometric modeling, gravity models, land value, urban economics, land use change, livelihood diversification, real estate management, urban permeability, prospect theory, and social urban theory.	PhD Candidate at Tohoku University, Japan
					1- Microsimulation methods have focused in modeling urban areas, and have not addressed post-disaster mobility in rural areas.					
4	Data Warehouse	N/A	Data Base	1- Earth system data extracted through a standard methodological process enables cross-disciplinary analytics to inform decision-making on-demand.	1- First global data cube with environmental (remote sensing), socioeconomic (land use and transport), demographic (fine-grained population), and urban/land use, bld. height, bld. volume, bld. footprint, household size, distribution, etc.) data.	✓			Senior data scientist, data engineering, model formulation and development, deployment, and problem-solving, Python, data structures, compression, cloud storage, instalment, processing, and analytics.	PhD Candidate at Tohoku University, Japan
5	Community Redevelopment	N/A	Recovery Framework	1- A standardized (ISO-seeking) temporal settlement to guide community-driven neighborhood and housing design and facilitate land ownership, home-business entrepreneurs, permits, and building codes compliance.	1- A neighborhood conformation algorithm using micro-data to enable small industry and service businesses compatible with the region.		✓		Community development planning, neighborhood design, resilience infrastructure design, household allocation, land use change evaluation, travel efficiency, and assisted housing design.	Program Manager at One United Recovery, Japan
6	Resilience Evaluation Framework	N/A	Resilience Framework	1- This framework consolidates 5 psychological evaluation tools to assess a mutual-help housing reconstruction process as a psychological intervention to evaluate whether disaster victims experience mental health effects that uplift their perceived self-efficacy, self-esteem, resilience, and satisfaction to speed livelihood recovery.	1- Mutual-help reconstruction has never been applied as a therapeutical intervention for community resilience post-disaster. 2- The selected evaluation tools have never been fused to develop a disaster resilience evaluation framework.		✓		Mental health data monitoring, analysis and evaluation research in humanitarian and disaster scenarios, psychosocial development, cross-sector program development, and multidisciplinary team leadership.	Program Manager at One United Recovery, Japan
7	Home-Kits	N/A	Product	1- A set of 3 unique structural components with unlimited flat-pack configurations for bottom-of-the-pyramid housing and a myriad of other applications.				✓	Mentorship and workshop management, product design, mechanical engineering, lean start up, entrepreneurship, business model development, stakeholder engagement, R&D innovation, industrial engineering.	Program Manager at Martin Garcia Fry & Associates
8	Modular Construction System	Housing	Program	1- Mainstreams hazard-resistant building codes in vulnerable high-risk areas, providing low income families access to affordable and safe housing.	1- First shelter prototype to guide people with no previous knowledge of construction into permanent housing, achieving engineered self-help housing.			✓	Building science, incremental construction, building codes, structural engineering, seismic-resistance, patent document drafting, patent search, technical innovation, collaboration, and partnership development.	Program Manager at One United Recovery, Japan
9	Carbon-fiber reinforced biocomposite polymer	Bioresin	Product	1- Formulated 5 biocomposite matrices made of softwood components to develop a scalable and biodegradable material for high-performance engineering.	1- This product is 100% softwood-based and depends on renewable material sources, making it possible to sustain continuous extraction with a very low carbon footprint 2- Based on the compact approach to material sourcing, production could be remote and linked to major supply chain networks. 3- Possibility of replacing the plastic industry.		✓		Biochemistry, environmental science, ecological footprint quantification, circular economies, waste management, materials engineering, structural homogeneity, injection molding, Thermoplastic Starch, reinforced polymers, carbon fiber manufacturing, research, applied science.	Program Manager at One United Recovery, Japan
10	Neighborhood-Driven Livelihood Recovery	Cyclical Leadership Model	Program	1- Applied psychographic segmentations, job-based land use, and household distribution data to consolidate a diversified neighborhood cooperative, and simulate a cyclical leadership model for a small-holder dairy farming business, optimizing net profits by 12% against elected business leaders.	1- These data sources have not been combined to estimate household integration for economic community development.		✓		Social science, social capital, financial modeling, econometrics, business model optimization, dairy farming sustainable practices, micro-data surveys, algorithm development, micro data assimilation, evaluation, and economic savings.	Masters Program at Tohoku University, Japan.
11	Smart Home Panel	Communications	Product	1- Robust monitoring and decision-making instrument to lead state-wide efforts. This product optimizes real-time decision-making, helps to monitor pre- and post-disaster-prone communities, inform on weather changes, alleviate poverty, and handle government surveys to inform policy development.	1- Direct government and household communication network, facilitating emergency assistance, optimizing household energy consumption, and administering strategic communications.	✓			Public sector stakeholder engagement, needs assessment, research, strategy development, collaborative design and development, sensor network, and IoT-Arduino programming.	Program Manager at One United Recovery, Japan
12	Solar-powered electromagnetic generator	Renewable Energy	Product	1- A/C electromagnetic generator derived from existing model to overcome the second law of thermodynamics, limiting every action with an equally responsive reaction, by using solar energy to generate continuous momentum.	1- How this can be achieved is only hypothesized (under R&D).	✓			Physics, thermodynamics, renewable energy production, electrical system engineering, mechanical engineering, and photovoltaic energy.	Program Manager at One United Recovery, Japan
13	Agronomic Urban Boundary Networks	AUB Networks	Vision	1- Novel urban growth management strategy to transition the intensity of urban growth and consolidate agricultural land to sustain the environmental stability of rural areas, while providing access to urban services and resilient infrastructure. This incremental development solution contributes to science and humanity by decarbonizing regions, administering environmental resources, and consolidating an urban-rural belt to protect rural areas from resource-intensive consumption and urban areas from horizontal disaster risks (volcanic eruptions, tornadoes, tsunamis, sea level rise, and floods).	1- First urban growth planning strategy to propose a whole-of-systems approach to mitigate climate change and enhance Human-Earth life support systems.		✓		Urban and regional planning, macroeconomics, carbon accounting, systems thinking, central place theory, graph theory, systems theory, growth pole theory, real estate management, transportation, accessibility, and econometric land use modeling.	PhD and Master's Degree Candidate at Tohoku University, Japan.